

MODULE 1 · FOUNDATIONS OF BIOINFORMATICS & BIOLOGICAL DATA

Molecular Biology Foundations

Lecture 2 :

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Agenda

- 1 The central dogma: DNA → RNA → Protein
- 2 Comparing DNA, RNA, and protein structure
- 3 Transcription and translation, step by step
- 4 Gene structure and basic regulation
- 5 Activities: diagram the dogma + trace a mutation
- 6 Practical: transcribe and translate with Biopython

BY THE END OF THIS LECTURE, YOU WILL BE ABLE TO

Learning Objectives



Explain the central dogma of molecular biology



Differentiate DNA, RNA, and protein structure



Describe basic gene structure and regulation

The Central Dogma at a Glance



DNA is transcribed into messenger RNA in the nucleus, then mRNA is translated into a chain of amino acids — a protein — at the ribosome.

DNA, RNA, and Protein Structure



DNA

- Sugar: deoxyribose
- Bases: A, T, G, C
- Double-stranded helix
- Stable, long-term storage



RNA

- Sugar: ribose
- Bases: A, U, G, C
- Usually single-stranded
- Short-lived messenger



Protein

- Building blocks: 20 amino acids
- Folds into 3D structure
- Determined by codon sequence
- Performs cellular function

Transcription & Translation, Step by Step



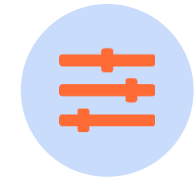
Transcription (nucleus)

- 1 RNA polymerase binds the gene's promoter
- 2 DNA strands separate locally
- 3 RNA polymerase synthesizes mRNA from the template strand
- 4 Pre-mRNA is processed (capped, spliced, polyadenylated)



Translation (ribosome)

- 1 Ribosome assembles on mature mRNA
- 2 tRNAs bring amino acids matching each codon
- 3 Ribosome links amino acids into a growing chain
- 4 Chain is released and folds into a functional protein



Gene Structure & Regulation



A gene includes regulatory and untranslated regions, not just coding sequence.

Gene regulation controls when and how much a gene is expressed:

- Transcription factors bind promoters and enhancers to turn genes on or off
- Enhancers/silencers can act from a distance via DNA looping
- Epigenetic marks (methylation, chromatin state) further modulate access

Diagram the Central Dogma

In small groups, take a given gene sequence and diagram it end-to-end:

1. Identify the template strand and transcribe it into mRNA
2. Mark the reading frame and translate it into amino acids
3. Label each stage: DNA → pre-mRNA → mature mRNA → protein

Each group presents their diagram to the class.



Case Study: Tracing a Point Mutation

Starting from a single base change in a DNA sequence, trace its effect forward:

- Does it change the mRNA codon?
- Is the resulting amino acid different (missense), the same (silent), or a stop (nonsense)?
- What might that mean for protein function?

Discuss as a class: why do some mutations matter more than others?



Practical: Transcribe & Translate with Biopython



Use Biopython's Seq object to transcribe and translate a DNA sequence programmatically.

```
from Bio.Seq import Seq

coding_dna = Seq("ATGGCCATTGTAATGGGCCGCTGAAAGGGTGCCCGATAG")

mrna = coding_dna.transcribe()
print("mRNA: ", mrna)

protein = coding_dna.translate()
print("Protein:", protein)
```

Assignment



Given a DNA sequence, manually derive the mRNA and protein — then verify your answer with code.

Show your work: the template strand, the transcribed mRNA, the reading frame, and the translated amino acid sequence — alongside the Biopython output.



Quick Check for Understanding

Short quiz — put these steps of the central dogma in the correct order:

Protein folds into its functional shape

RNA polymerase transcribes the gene into pre-mRNA

Mature mRNA leaves the nucleus for the ribosome

Ribosome translates mRNA codons into amino acids

COMING UP NEXT

Lecture 3: Biological Databases

Navigating NCBI, Ensembl, UniProt, and PDB — with a Biopython Entrez practical for programmatic record retrieval.

Bring: your assignment showing manual + code-verified transcription/translation.

